

NEET Important Chapter – Gravitation

Gravitational Force and Newton's Law of Gravitation: An Important

Search your topic here...

The gravitation chapter covers the most important concepts for [NEET](#), JEE and other competitive exams. In the Gravitation chapter, students will study the basic gravitation definition and Newton's law of gravitation.

The chapter on gravitation begins with the very basic definition of gravity, what is gravitation and Newton's law of gravitation. Further, the chapter covers the concept of acceleration due to gravity and its variations with different parameters. These concepts will lead us to a very important part of the chapter i.e., planetary motion and Kepler's law. In this chapter, we will get to study many interesting and most important concepts such as gravitational constant, gravitational potential, satellites and terms related to satellites, weightlessness and Binding energy.

The concept of the gravitational field will help us in studying the gravitational potential and potential energy where we can see the resemblance with the electrostatic fields. In this article, we will solve many important numerical problems that will help us in acing the exams.

In this article, we will cover the important concepts and topics that will help students to revise and boost their preparations for NEET/JEE.

Important Topics of Gravitation

- Universal law of gravitation
- Acceleration due to gravity
- Variations of acceleration due to gravity
- Kepler's laws of planetary motion
- Gravitation potential and gravitational potential energy
- Satellites
- Binding energy
- Weightlessness

Important Concepts of Gravitation

Sl.No	Name of the Concept	Key Points
1.	Universal Law of Gravitation	• This explains that gravitational force is an attractive force. We experience many gravitation examples in our day to day lives. • Newton's universal law of gravitation or Newton's gravitational law states, "The force of attraction between two masses will be directly proportional to the product of masses and inversely proportional to the square of the distance between them."

2.	Acceleration due to gravity	• It is the acceleration of the body which is subjected to free fall. It is denoted by the letter g. • It is independent of the size, shape and mass of the body. • The value of g on earth is 9.8 m/s^2.
3.	Kepler's first law of planetary motion	• The first law states, "All the planets in our solar system are revolving around an elliptical orbit with the sun at one of the foci".
4.	Kepler's second law of planetary motion	• The second law states, "The rate of the total area swept by the area vector of the revolving planet with respect to the Sun per unit time remains unchanged irrespective of the position of the planets on the set elliptical path." i.e., — • Kepler's second law follows conservation of angular momentum. (image will be uploaded soon)
5.	Kepler's third law of planetary motion	• The third law of planetary motion states, "The square of the period of revolution around the sun of a planet is proportional to the cube of the semimajor axis of its orbit-path around the sun" i.e.,

6.	Types of satellites	• Geostationary Satellite: A geostationary satellite is one that circles around the planet in its equatorial plane at the same angular speed and in the same direction as the earth spins around its own axis. • Polar satellite: A polar satellite is a satellite that circles in a polar orbit. • Natural Satellite: A natural satellite is a celestial body that circles around a planet. The moon, for example, is a natural satellite of the earth. (image will be uploaded soon)
7.	Weightlessness	• The sensation of weightlessness occurs when the body is unsupported and no external force is acting on it. • Because there is no external force acting on an object in free fall with an acceleration equal to that of the earth, it is said to be weightless. • For example, when we are travelling through a lift we experience weightlessness.

List of Important Formulae of Gravitation

Sl. No	Name of the Concept	Formulae
--------	---------------------	----------

1.	Newton's universal law of gravitation	_____ _____ Where, G- The universal gravitational constant and its value is
2.	Acceleration due to gravity of earth	_____ Where, M_e - The mass of the earth R_e - The radius of the earth

3.	Variation of acceleration due to gravity with height (h)	
4.	Variation of acceleration due to gravity with depth (d)	—
5.	Variation of acceleration due to gravity with rotation of the earth about its axis	Note: a. At the equator, _____, value acceleration due to gravity b. At the pole, _____, value acceleration due to gravity

6.	Gravitational field and potential energy	_____ _____
7.	Escape velocity of satellites	_____ _____
		_____ _____

8.	Orbital velocity of satellites	_____ When the satellite is orbiting close to the earth's surface, $h \ll R_e$, then: _____
9.	Time period of satellite	_____
10.	Total energy of the satellite	_____
11.	Binding Energy	_____

Important Questions of Gravitation

1. The change in the value of g at a height h above the surface of the earth is the same as at a depth d below the surface of earth. When both d and h are much smaller than the radius of earth, then which of the following is correct?

- a. $d = 2h$
- b. $d = h$
- c. $d = -$
- d. $d = -$

Answer: (a)

$$\Rightarrow g\left(\frac{2h}{R}\right) = g\frac{d}{R}$$

$$\Rightarrow d = 2h$$

2. The average density of the earth?

- (a) is directly proportional to g
- (b) is inversely proportional to g
- (c) does not depend on g
- (d) is a complex function of g

Answer: (a)

$$\text{Average density of earth} = \frac{M}{\frac{4}{3}\pi R^3}$$

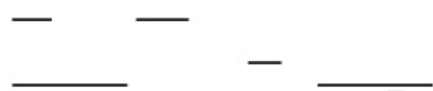
$$= \frac{gR^2}{\frac{4}{3}\pi \times \frac{3}{gR^2}} = \frac{3g}{4\pi R}$$

So, Average density of the earth is directly proportional to g .

3. A satellite of the sun is in a circular orbit around the sun, midway between the sun and the earth. Then

- (a) The period of the satellite is nearly 229 days**
- (b) The period of the satellite is nearly 129 days**
- (c) The speed of the satellite equal the escape velocity of the earth.**
- (d) The acceleration of the satellite is four times the acceleration of the Earth.**

Answer: (b)



4. When you move from equator to pole, the value of acceleration due to gravity (g)

- a. Increases**
- b. Decreases**
- c. Remains the same**
- d. Increases then decreases**

Answer: (a)

From equation:

Where θ is the latitude which is 0 at the equator and 90 at the poles. Therefore g is maximum at poles and minimum at equator. Hence value of g increases while going from the equator poles.

5. The weight of the body at the centre of the earth is

- a. Zero**
- b. Infinite**
- c. Same as on the surface**
- d. None of these**

Answer: (a)

Because g at centre of the earth is zero so weight = mg is also zero.

Solved Examples of Gravitation

1. An artificial satellite was moving in a circular orbit such that its total energy was $2E_0$. Then, calculate the potential energy of the given satellite.

- a. E_0
- b. $3 E_0$
- c. $4 E_0$
- d. $-E_0$

Sol:

Given,

The total energy of the satellite $= E = 2 E_0$

We know that the total energy of the satellite is given by= _____

Since, $R_e \gg h$, we get:

Now, the potential energy of the satellite is given by:

Therefore, the potential energy of the satellite is $4 E_0$. Hence, the right option is "c".

Key Point: The relationship between the total energy and potential energy of the satellite is $U=2E$.

2. The height at which the acceleration due to gravity becomes $\frac{1}{4}$ in terms of the radius of the earth R , is

a. $\frac{R}{2}$

b. $2R$

c. $\frac{R}{4}$

d. $3R$

Sol:

Given,

Acceleration due to gravity at a height " h " = $g_h = \frac{g}{4}$

Now, we are asked to determine the height h in terms of radius of earth. We know that, variation of acceleration due to gravity with height is given by:

$$\frac{g_h}{g} = \left(\frac{R}{R+h} \right)^2$$

$$\frac{1}{4} = \left(\frac{R}{R+h} \right)^2$$

$$\frac{1}{4} = \frac{R^2}{(R+h)^2}$$

Therefore, option "d" is the right answer.

Key point: Here, the formula of acceleration due to gravity with the variation of height can be used directly.

Previous Year Questions of Gravitation

1. Two stars of masses m and $2m$ at a distance d rotate about their common center of mass in free space. The period of revolution is: (JEE 2021)

a.

b.

c.

d.

Sol:

Given,

Two stars of masses m and $2m$ are placed at a distance of d . Now, it is being said that they are rotating about their own axis in free space. We know that these two stars are able to rotate in a circular motion due to the force of attraction (gravitational attraction). Then we need to determine the center of mass for these two distances.

We know that, center of mass can be calculated by using below equation:

Let us assume that we are considering m as the initial position of the system then we get:

So, now the **force due to gravity** acting between the two stars will be equal to the centripetal force acting towards the C.O.M. We write:

Therefore, option A is the right answer.

Trick: The C.O.M is considered as a static point, thus the masses will always rotate about the same axis.

2. Consider two solid spheres of radii $R_1 = 1m$, $R_2 = 2m$ and masses M_1 and M_2 , respectively. The gravitational field due to sphere (1) and (2) are shown. The value of — is : (JEE main 2020)

a. —

b

a. ☐

c. ☐

d. ☐

Sol:

Given,

The radii of the two spheres are $R_1 = 1m$, $R_2 = 2m$ and masses M_1 and M_2 , respectively. We are asked to determine the ratio of the masses. From the gravitational field equation, we know that:

For given two spheres we get:

The ratio of masses will be:

Therefore, the right answer is option b.

Trick: We used the graphs to know the value of gravitational fields instead of using the formula for determining the value of the gravitational field.

Practice Questions

1. A straight rod of length L extends from $x = a$ to $x = L + a$. The gravitational force it exerts on a point mass ' m ' at $x = 0$, if the mass per unit length of the rod is $A + Bx^2$, is given by?

Ans:

— —

2. If the Earth has no rotational motion, the weight of a person on the equator is W . Determine the speed with which the earth would have to rotate about its axis so that the person at the equator will weigh $\frac{1}{2}W$. Radius of the Earth is 6400 km and $g=10 \text{ m/s}^2$.

Ans:

Conclusion

We conclude that the chapter gravitation has many important concepts that will help us in scoring better marks in NEET and JEE. In this article, we covered gravitation introduction, principles of gravity and all other important concepts from the chapter along with solved and previous year questions. Students can test their knowledge with the help of practice questions.

Hi Aditya,

Best courses for you

**Full syllabus LIVE
courses**

Starting from ₹32,400

**One-to-one LIVE
classes**

Starting from ₹0/hr

Watch videos on
NEET Important Chapter - Gravitation

Gravitation Class 11 | One Shot | Marathon | JEE Main | JEE Advanced | Vinay Shur Sir | VJEE

Subscribe

Share

3.2K likes 122.6K Views 3 years ago

Book your **Free Demo** session

Get a flavour of LIVE classes here at Vedantu

[Book a free demo](#)

Vedantu Improvement Promise

We promise improvement in
marks or get your fees
back. T&C Apply*